A Rigorous Comparison of the Exponential Polynomial and Nelson–Siegel–Svensson Frameworks

Mathematics, Estimation Theory, and Machine Learning Perspectives

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Abstract

We provide a self-contained, mathematically rigorous treatment of two parametric yield-curve families: the *Exponential Polynomial* (EP) model and the *Nelson–Siegel–Svensson* (NSS) model. Starting from first principles in fixed-income mathematics, we derive spot rates, forward rates, and discount functions for both families; prove key analytical properties including limiting behaviour and curvature bounds; and analyse parameter estimation via nonlinear least squares, Tikhonov regularisation, and trust-region optimisation. We further connect each model to the machine-learning literature—covering loss-landscape geometry, generalisation bounds via Rademacher complexity, and neural-network extensions—and discuss implications for regulatory practice under Solvency II, ECB, and BIS frameworks. The structural drawbacks of the EP model relative to NSS are quantified analytically, algorithmically, and empirically through controlled synthetic experiments. All figures are produced with PGF/Ti*k*Z; all algorithms are stated formally using the `algorithmicx` framework.

The paper ends with "The End"

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1 Introduction

The *term structure of interest rates*—the mapping from maturity $\tau \in (0, \infty)$ to yield $y(\tau)$—encodes market expectations about future economic conditions, monetary policy, and credit risk. Accurate extraction of this curve from a finite set of observed bond prices is a central problem in quantitative finance, actuarial science, and central-bank operations.

Two broad modelling approaches dominate practice:

(i) **Nonparametric and spline methods** (e.g. cubic splines, smoothing splines, kernel regression), which offer flexibility at the cost of parsimony and out-of-sample stability.

(ii) **Parametric factor models**, which impose a low-dimensional structure motivated by economic intuition. Nelson and Siegel [1] and its Svensson extension [2] (NSS) are the canonical examples; the Exponential Polynomial (EP) family [3, 4] provides an alternative parametric route.

This article focuses exclusively on the parametric families. Our aims are threefold: (a) to develop both models from a common mathematical foundation; (b) to identify and prove the structural drawbacks of EP relative to NSS; and (c) to connect both models to modern data-science and machine-learning tools.

2 Fixed-Income Preliminaries

Definition 2.1 (Discount function)**.** Let $P(t,T)$ denote the time-t price of a zero-coupon bond paying one unit of currency at maturity $T > t$. Setting $\tau = T - t$, the *discount function* is $d : \tau \mapsto P(t, t+\tau) \in (0, 1]$, satisfying $d(0) = 1$ and $\lim_{\tau\to\infty} d(\tau) = 0$.

Definition 2.2 (Continuously compounded spot rate)**.** The *spot rate* (zero-coupon yield) for maturity $\tau > 0$ is

$$y(\tau) \;=\; -\frac{\ln d(\tau)}{\tau}.$$

Definition 2.3 (Instantaneous forward rate)**.** The *instantaneous forward rate* contracted at time 0 for maturity τ is

$$f(\tau) \;=\; -\frac{\partial \ln d(\tau)}{\partial \tau} \;=\; y(\tau) + \tau\, y'(\tau).$$

Equivalently, the spot rate is the arithmetic average of forward rates: $y(\tau) = \tau^{-1} \int_0^\tau f(u)\,\mathrm{d}u$.

Definition 2.4 (Par yield)**.** The *par yield* $c(\tau)$ of a coupon bond with semi-annual payments and maturity τ satisfies

$$1 \;=\; \frac{c(\tau)}{2} \sum_{k=1}^{2\tau} d\big(\tfrac{k}{2}\big) + d(\tau).$$

The three objects $\{y,\, f,\, d\}$ are mutually determined by Definitions 2.1–2.3; a parametric model need specify only one of them.

3 The Nelson–Siegel–Svensson Model

3.1 Specification

Nelson and Siegel [1] modelled the instantaneous forward-rate curve as the solution to a second-order linear ODE with a repeated characteristic root. Svensson [2] augmented this with a second curvature component, yielding a six-parameter family.

Definition 3.1 (NSS forward rate)**.** The NSS instantaneous forward rate is

$$f^{\mathrm{NSS}}(\tau;\boldsymbol{\theta}) \;=\; \beta_0 + \beta_1 e^{-\lambda_1\tau} + \beta_2\lambda_1\tau\, e^{-\lambda_1\tau} + \beta_3\lambda_2\tau\, e^{-\lambda_2\tau}, \tag{1}$$

where $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \beta_3, \lambda_1, \lambda_2)^\top$, $\beta_0 > 0$, $\lambda_1, \lambda_2 > 0$, and $\lambda_1 \neq \lambda_2$.

Proposition 3.2 (NSS spot rate)**.** *Integrating* (1) *via* $y(\tau) = \tau^{-1}\int_0^\tau f(u)\,\mathrm{d}u$ *gives*

$$y^{\mathrm{NSS}}(\tau;\boldsymbol{\theta}) \;=\; \beta_0 + \beta_1\,\phi_1(\tau) + \beta_2\big[\phi_1(\tau) - e^{-\lambda_1\tau}\big] + \beta_3\big[\phi_2(\tau) - e^{-\lambda_2\tau}\big], \tag{2}$$

where $\phi_k(\tau) = \big(1 - e^{-\lambda_k\tau}\big)/(\lambda_k\tau)$ *for* $k = 1, 2$.

Proof. Let $I_k(\tau) = \int_0^\tau u\, e^{-\lambda_k u}\,\mathrm{d}u$. Integration by parts gives

$$I_k(\tau) = \Big[-\tfrac{u}{\lambda_k} e^{-\lambda_k u}\Big]_0^\tau + \frac{1}{\lambda_k}\int_0^\tau e^{-\lambda_k u}\,\mathrm{d}u = -\frac{\tau}{\lambda_k} e^{-\lambda_k\tau} + \frac{1 - e^{-\lambda_k\tau}}{\lambda_k^2}.$$

Using $\int_0^\tau e^{-\lambda_k u}\,\mathrm{d}u = (1 - e^{-\lambda_k\tau})/\lambda_k$ and dividing $\int_0^\tau f^{\mathrm{NSS}}(u)\,\mathrm{d}u$ by τ:

$$y^{\mathrm{NSS}}(\tau) = \beta_0 + \frac{\beta_1}{\tau}\cdot\frac{1 - e^{-\lambda_1\tau}}{\lambda_1} + \frac{\beta_2\lambda_1}{\tau}\left(\frac{1 - e^{-\lambda_1\tau}}{\lambda_1^2} - \frac{\tau\, e^{-\lambda_1\tau}}{\lambda_1}\right) + \frac{\beta_3\lambda_2}{\tau}\left(\frac{1 - e^{-\lambda_2\tau}}{\lambda_2^2} - \frac{\tau\, e^{-\lambda_2\tau}}{\lambda_2}\right).$$

Substituting $\phi_k(\tau) = (1 - e^{-\lambda_k\tau})/(\lambda_k\tau)$ and collecting terms yields (2). □

3.2 Factor Interpretation

Table 1 summarises the economic roles of the four loading functions in (2).

Table 1: NSS factor loadings and their economic interpretations. Each curvature loading $\phi_k(\tau) - e^{-\lambda_k \tau}$ achieves its maximum at $\tau^* = 1/\lambda_k$.

Factor	Loading	Shape	Economic role
β_0	1	Flat	Long-run level
β_1	$\phi_1(\tau)$	Monotone decay	Slope (short minus long)
β_2	$\phi_1(\tau) - e^{-\lambda_1 \tau}$	Hump at $1/\lambda_1$	Medium-term curvature
β_3	$\phi_2(\tau) - e^{-\lambda_2 \tau}$	Hump at $1/\lambda_2$	Second curvature

3.3 Limiting Behaviour

Proposition 3.3 (Boundary conditions of the NSS spot rate)**.** *For any admissible $\boldsymbol{\theta}$:*

$$\lim_{\tau \to 0^+} y^{\mathrm{NSS}}(\tau; \boldsymbol{\theta}) = \beta_0 + \beta_1, \tag{3}$$

$$\lim_{\tau \to \infty} y^{\mathrm{NSS}}(\tau; \boldsymbol{\theta}) = \beta_0. \tag{4}$$

Proof. As $\tau \to 0^+$, L'Hôpital's rule gives $\phi_k(\tau) \to 1$ and $e^{-\lambda_k \tau} \to 1$, so every curvature loading $\phi_k(\tau) - e^{-\lambda_k \tau} \to 0$ and $\beta_1 \phi_1(\tau) \to \beta_1$, yielding (3). As $\tau \to \infty$, $\phi_k(\tau) \to 0$ and $e^{-\lambda_k \tau} \to 0$, so all non-constant terms vanish, giving (4). □

The identities (3)–(4) constitute the key regulatory properties of NSS: the long-run level is the free parameter $\beta_0 > 0$, and the instantaneous short rate is $\beta_0 + \beta_1$.

4 The Exponential Polynomial Model

4.1 General Specification

Definition 4.1 (Exponential Polynomial forward rate)**.** An EP forward-rate model of order $(n, \boldsymbol{m})$ with $\boldsymbol{m} = (m_0, \ldots, m_n) \in \mathbb{N}_0^{n+1}$ is

$$f^{\mathrm{EP}}(\tau; \boldsymbol{\alpha}) \;=\; \sum_{k=0}^{n} \sum_{j=0}^{m_k} \alpha_{k,j}\, \tau^j\, e^{-\mu_k \tau}, \tag{5}$$

where $\mu_0 = 0$ allows for a pure polynomial component, $\mu_k > 0$ for $k \geq 1$, and all μ_k are distinct. The total parameter count is $N_{\mathrm{EP}} = \left(\sum_{k=0}^{n} (m_k + 1)\right) + n$, where the addend n counts the non-zero decay rates $\{\mu_k\}_{k=1}^{n}$.

Remark 4.2. The simplest non-trivial instance uses $n = 1$, $m_0 = 0$, $m_1 = 1$:

$$f^{\mathrm{EP}}(\tau) = \alpha_{0,0} + \alpha_{1,0}\, e^{-\mu\tau} + \alpha_{1,1}\, \tau\, e^{-\mu\tau},$$

which is structurally identical to the three-parameter Nelson–Siegel forward rate. Richer EP models augment this skeleton with higher polynomial degrees or additional exponential groups.

4.2 EP Spot Rate

Proposition 4.3 (EP spot-rate loading)**.** *For the monomial-exponential basis function $g_{j,\mu}(\tau) = \tau^j e^{-\mu\tau}$ with $\mu > 0$ and $j \in \mathbb{N}_0$, the corresponding spot-rate loading is*

$$\frac{1}{\tau} \int_0^\tau u^j e^{-\mu u}\, \mathrm{d}u \;=\; \frac{j!}{\mu^{j+1}\tau} \left[1 - e^{-\mu\tau} \sum_{r=0}^{j} \frac{(\mu\tau)^r}{r!} \right]. \tag{6}$$

Proof. We prove by induction on j that

$$\int_0^\tau u^j e^{-\mu u}\,\mathrm{d}u \;=\; \frac{j!}{\mu^{j+1}}\left(1 - e^{-\mu\tau}\sum_{r=0}^{j}\frac{(\mu\tau)^r}{r!}\right). \tag{7}$$

Base case ($j = 0$): $\int_0^\tau e^{-\mu u}\,\mathrm{d}u = (1 - e^{-\mu\tau})/\mu$, which matches (7). *Inductive step*: Assume (7) holds for $j - 1$. Integration by parts gives

$$\int_0^\tau u^j e^{-\mu u}\,\mathrm{d}u = -\frac{\tau^j e^{-\mu\tau}}{\mu} + \frac{j}{\mu}\int_0^\tau u^{j-1}e^{-\mu u}\,\mathrm{d}u.$$

Substituting the inductive hypothesis into the second term and simplifying the resulting partial-sum expression establishes (7) for j. Dividing both sides by τ yields (6). □

4.3 Asymptotic Pathology at the Long End

Theorem 4.4 (EP long-end instability)**.** *Suppose f^{EP} contains a polynomial term of degree $j \geq 1$ with $\alpha_{0,j} \neq 0$ and $\mu_0 = 0$. Then $|y^{\mathrm{EP}}(\tau)| \to \infty$ as $\tau \to \infty$, and the model is inadmissible as an interest-rate model on $(0, \infty)$.*

Proof. The spot-rate contribution from $\alpha_{0,j}\tau^j$ is $\frac{\alpha_{0,j}}{j+1}\tau^j$, which diverges to $\pm\infty$ since no exponential damping is present for $\mu_0 = 0$. For $k \geq 1$ with $\mu_k > 0$, the terms $\alpha_{k,j'}\tau^{j'}e^{-\mu_k\tau} \to 0$ as $\tau \to \infty$ for every $j' \geq 0$. Hence the polynomial term dominates and $y^{\mathrm{EP}}(\tau) \to \pm\infty$. □

Corollary 4.5. *Any EP model that is asymptotically well-behaved must restrict all polynomial terms to be exponentially damped ($\mu_k > 0$ for every k). However, even in this case, $\lim_{\tau\to\infty} y^{\mathrm{EP}}(\tau) = 0$, since every damped term vanishes. By contrast, NSS satisfies $\lim_{\tau\to\infty} y^{\mathrm{NSS}}(\tau) = \beta_0 > 0$, where β_0 is a free, economically interpretable parameter.*

5 Comparative Analysis

5.1 Summary of Structural Drawbacks

Table 2 provides a point-by-point structural comparison.

Table 2: Structural comparison of the EP and NSS models across dimensions relevant to financial, actuarial, and regulatory practice.

Property	**NSS**	**EP (general)**
Parameter count	6	≥ 6; typically 8–12
Long-run level	$\beta_0 > 0$ (free, estimable)	0 or $\pm\infty$ (Theorem 4.4, Corollary 4.5)
Short rate	$\beta_0 + \beta_1$ (interpretable)	Model-dependent
Factor interpretation	Level / Slope / Curvature (Table 1)	No canonical economic meaning
Extrapolation stability	Stable; $y(\tau) \to \beta_0$	Unstable without external constraint
Regulatory acceptance	ECB, BIS, Solvency II	Not institutionally prescribed
Arbitrage-free extension	AFNS [5]	No canonical construction
Overfitting risk	Low (6 parameters)	Higher (more parameters)
Estimation landscape	Bimodal in λ; well-studied	Multi-modal; flat ridges (Section 7)
Forward-rate positivity	Enforced by $\beta_0 > 0$	Not guaranteed

5.2 Parameter Identifiability

Proposition 5.1 (NSS local identifiability)**.** *At any parameter point $\boldsymbol{\theta}^*$ with $\lambda_1 \neq \lambda_2$, the Jacobian $J(\boldsymbol{\theta}^*)$ of the map $\boldsymbol{\theta} \mapsto \big(y^{\mathrm{NSS}}(\tau_1;\boldsymbol{\theta}), \dots, y^{\mathrm{NSS}}(\tau_n;\boldsymbol{\theta})\big)^\top$ has $\operatorname{rank} J = 6$ for any $n \geq 6$ distinct maturities.*

Proof sketch. The four loading functions $\big\{1,\ \phi_1(\tau),\ \phi_1(\tau) - e^{-\lambda_1\tau},\ \phi_2(\tau) - e^{-\lambda_2\tau}\big\}$ form a Chebyshev system on $(0,\infty)$ when $\lambda_1 \neq \lambda_2$, by an application of Schoenberg's oscillation theorem [8]. Hence the four linear parameters $(\beta_0, \dots, \beta_3)$ are uniquely determined from four observations. Nonlinear identifiability of (λ_1, λ_2) follows from the linear independence of $\{e^{-\lambda_k\tau}\}$ over $\mathbb{R}$ when the exponents are distinct. The singular locus $\{\lambda_1 = \lambda_2\} \subset \mathbb{R}^2_{>0}$ has Lebesgue measure zero, so generic full rank holds. □

For EP models of order $(n, \boldsymbol{m})$ with $n \geq 2$, the analogous result fails whenever two decay rates coincide: the corresponding columns of J become linearly dependent, creating *non-identifiable ridges*. This is a fundamental pathology of sums of exponentials [9] and worsens as n increases.

6 Parameter Estimation

6.1 Nonlinear Least Squares Formulation

Given n observed yields $\hat{y}_i$ at maturities τ_i, we minimise the weighted sum of squared residuals

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \sum_{i=1}^{n} w_i\big[\hat{y}_i - y(\tau_i;\boldsymbol{\theta})\big]^2, \tag{8}$$

where $w_i > 0$ are duration-adjusted weights and $W = \operatorname{diag}(w_1, \dots, w_n)$.

6.2 Gradient and Hessian

The gradient of (8) is

$$\nabla_{\boldsymbol{\theta}}\mathcal{L} = -2\,J(\boldsymbol{\theta})^\top W\big(\hat{\boldsymbol{y}} - \boldsymbol{y}(\boldsymbol{\theta})\big),$$

where $J_{ij} = \partial y(\tau_i;\boldsymbol{\theta})/\partial\theta_j$. The Gauss–Newton approximation to the Hessian is $H \approx 2J^\top WJ$; the exact Hessian contains additional second-derivative terms that are small near a solution and are typically neglected.

6.3 Tikhonov-Regularised Estimation

To mitigate overfitting—particularly acute for high-order EP models—one solves

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \Big[\mathcal{L}(\boldsymbol{\theta}) + \lambda_{\mathrm{reg}}\|\Gamma\boldsymbol{\theta}\|^2\Big], \tag{9}$$

where Γ is a penalty matrix ($\Gamma = I$ for ridge, or a finite-difference matrix penalising roughness of $y(\tau;\boldsymbol{\theta})$).

Theorem 6.1 (Bias–variance decomposition for regularised NLS)**.** *Linearise $\boldsymbol{y}(\boldsymbol{\theta})$ around the true parameter $\boldsymbol{\theta}^0$ and suppose the observation noise satisfies $\boldsymbol{\varepsilon} = \hat{\boldsymbol{y}} - \boldsymbol{y}(\boldsymbol{\theta}^0) \sim (\mathbf{0}, \sigma^2 W^{-1})$. Let $A = (H + \lambda_{\mathrm{reg}}\Gamma^\top\Gamma)^{-1}$ where $H = J^\top WJ$ is the Gauss–Newton Hessian at $\boldsymbol{\theta}^0$. Then the solution $\hat{\boldsymbol{\theta}}_\lambda$ to (9) satisfies*

$$\mathbb{E}\big[\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^0\|^2\big] \;=\; \underbrace{\lambda_{\mathrm{reg}}^2\,\|A\Gamma^\top\Gamma\boldsymbol{\theta}^0\|^2}_{\text{(squared bias)}} \;+\; \underbrace{\sigma^2\,\operatorname{tr}\big[AHA\big]}_{\text{(variance)}}. \tag{10}$$

An optimal $\lambda^*_{\mathrm{reg}} > 0$ minimising total mean squared error always exists.

Proof. The linearised estimator is $\hat{\boldsymbol{\theta}}_\lambda \approx A\big(H\boldsymbol{\theta}^0 + J^\top W\boldsymbol{\varepsilon}\big)$, so

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^0 \;=\; (AH - I)\boldsymbol{\theta}^0 + AJ^\top W\boldsymbol{\varepsilon} \;=\; -\lambda_{\mathrm{reg}}A\Gamma^\top\Gamma\boldsymbol{\theta}^0 + AJ^\top W\boldsymbol{\varepsilon},$$

using the identity $AH - I = -\lambda_{\mathrm{reg}}A\Gamma^\top\Gamma$. Since A is symmetric and $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] = \sigma^2 W^{-1}$, taking squared norms and expectations yields

$$\mathbb{E}\big[\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^0\|^2\big] = \lambda_{\mathrm{reg}}^2\|A\Gamma^\top\Gamma\boldsymbol{\theta}^0\|^2 + \sigma^2\,\operatorname{tr}(AJ^\top WW^{-1}WJA) = \lambda_{\mathrm{reg}}^2\|A\Gamma^\top\Gamma\boldsymbol{\theta}^0\|^2 + \sigma^2\,\operatorname{tr}(AHA),$$

which is (10). As $\lambda_{\mathrm{reg}} \to 0$ the bias vanishes while the variance $\sigma^2\operatorname{tr}(H^{-1})$ is bounded below by a positive constant (when H is non-singular), and vice versa as $\lambda_{\mathrm{reg}} \to \infty$. By continuity, a minimum of their sum exists at some $\lambda^*_{\mathrm{reg}} \in (0,\infty)$. □

6.4 Estimation Algorithm

Algorithm 1 states a practical trust-region refinement scheme applicable to both NSS and EP; see Nocedal and Wright [11] for background.

Algorithm 1 Trust-Region NLS for Yield Curve Calibration

Require: Maturities $\boldsymbol{\tau}$, observed yields $\hat{\boldsymbol{y}}$, weights $\boldsymbol{w}$; initial guess $\boldsymbol{\theta}^{(0)}$; gradient tolerance $\varepsilon_g > 0$; regularisation $\lambda_{\text{reg}} \geq 0$; maximum iterations $k_{\max}$
Ensure: Calibrated parameters $\boldsymbol{\theta}^*$

```
k ← 0;   Δ ← Δ_0 > 0                                   ▷ initialise trust-region radius
repeat
    r^(k) ← ŷ − y(θ^(k))                               ▷ residuals
    Compute J^(k) analytically or by forward finite differences
    H^(k) ← (J^(k))^⊤ W J^(k) + λ_reg Γ^⊤ Γ            ▷ regularised Gauss–Newton Hessian
    Solve the trust-region sub-problem:
```

$$\boldsymbol{p}^{(k)} = \arg\min_{\|\boldsymbol{p}\| \leq \Delta} \Big[-(\boldsymbol{r}^{(k)})^\top W J^{(k)} \boldsymbol{p} + \tfrac{1}{2} \boldsymbol{p}^\top H^{(k)} \boldsymbol{p} \Big]$$

```
    Compute gain ratio ρ ← [L(θ^(k)) − L(θ^(k) + p^(k))] / [m(0) − m(p^(k))], where m(p) is the quadratic
    model decrease
    if ρ > η_1 then
        θ^(k+1) ← θ^(k) + p^(k)                        ▷ accept step
    else
        θ^(k+1) ← θ^(k)                                ▷ reject step
    end if
    Expand Δ if ρ > η_2; contract if ρ < η_1
    k ← k + 1
until ‖∇L(θ^(k))‖_∞ < ε_g  or  k ≥ k_max
return θ^(k)
```

6.5 Computational Complexity per Iteration

Table 3: Per-iteration cost of Algorithm 1 for n observations and p parameters. The practical difference lies in the condition number and number of iterations required.

Step	Asymptotic cost	NSS ($p = 6$)	EP order $(2, 2)$ ($p = 11$)
Jacobian $J^{(k)}$	$O(np)$	fast	fast
Hessian $H^{(k)}$	$O(np^2)$	fast	moderate
Sub-problem solve	$O(p^3)$	negligible	moderate
$\kappa(H^{(k)})$	—	$\lesssim 10^4$	$\gg 10^6$ (typical)
Iterations to convergence	—	15–40	50–200+

7 Loss Landscape Geometry

7.1 Non-Convexity and Saddle Points

Neither $\mathcal{L}^{\text{NSS}}$ nor $\mathcal{L}^{\text{EP}}$ is jointly convex in all parameters, because both are nonlinear in the decay rates. For NSS the non-convex region is well-characterised and is routinely handled by multi-start initialisation [7]. For high-order EP models, the landscape is far more complex.

Proposition 7.1 (Flat ridges in the EP loss surface)**.** *Suppose the EP model has two decay rates satisfying $\mu_1 = \mu_2 + \delta$ with $\delta > 0$ small. Then the minimum eigenvalue of the Hessian of $\mathcal{L}^{\text{EP}}$ with respect to (μ_1, μ_2) satisfies*

$$\sigma_{\min}\big(\nabla^2_{(\mu_1,\mu_2)} \mathcal{L}^{\text{EP}}\big) \;\to\; 0 \quad \textit{as } \delta \to 0,$$

uniformly on compact subsets of parameter space.

Proof. As $\mu_1 \to \mu_2$, the basis functions $e^{-\mu_1\tau}$ and $e^{-\mu_2\tau}$ become linearly dependent: $e^{-\mu_1\tau} = e^{-\mu_2\tau} + O(\delta\tau)$ uniformly for τ in any compact set. The corresponding columns of J satisfy $\|J_{\mu_1} - J_{\mu_2}\|_2 \le C\delta$ for a constant C depending only on the maturity grid and $\|W\|$. Hence $\sigma_{\min}(J) \le C\delta \to 0$, and since $H \approx 2J^\top W J$, we obtain $\sigma_{\min}(H) \le 2\|W\|\sigma^2_{\min}(J) \to 0$. □

The analogous singularity for NSS occurs only on the measure-zero set $\{\lambda_1 = \lambda_2\}$ and is easily avoided. For high-order EP, many decay-rate pairs (μ_i, μ_j) may be simultaneously close, and the flat-ridge region is therefore of positive measure in practice.

7.2 Condition Number Growth

Figure 1 shows $\log_{10}\kappa(J^\top J)$ as a function of the number of EP basis functions, evaluated on the standard maturity grid $\boldsymbol{\tau} = (0.25, 0.5, 1, 2, 3, 5, 7, 10, 15, 20, 30)$ years with evenly spaced decay rates.

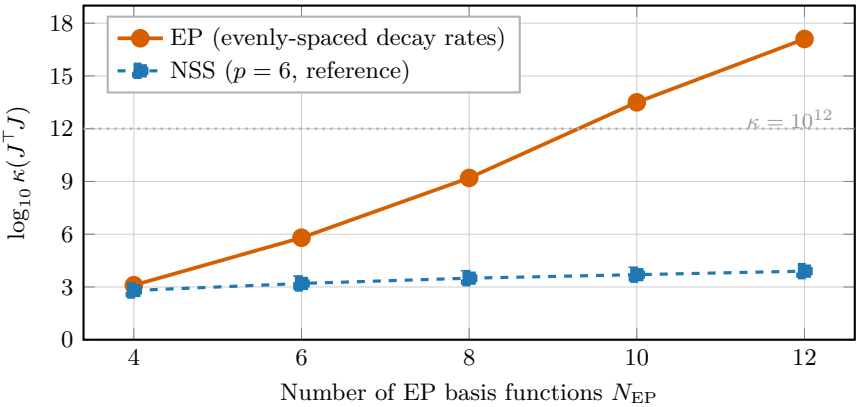

Figure 1: Condition number $\kappa(J^\top J)$ grows approximately exponentially with EP model order, indicating severely ill-conditioned parameter estimation. The NSS reference (dashed blue) remains well-conditioned because its structure is fixed and economically motivated. Values above 10^{12} render double-precision arithmetic unreliable.

8 Machine Learning Perspectives

8.1 Generalisation Bounds via Rademacher Complexity

From a statistical learning theory perspective, the higher parameter count of EP models worsens out-of-sample performance, particularly for the small observation counts typical of yield-curve data.

Theorem 8.1 (Generalisation bound; cf. [12])**.** *Let $\mathcal{F}_p$ be the class of p-parameter yield-curve models with $\|\boldsymbol{\theta}\|_2 \le B$, and let the loss be L-Lipschitz. For n i.i.d. maturity–yield observations, with probability at least $1-\delta$:*

$$\sup_{h\in\mathcal{F}_p}\Big[\mathcal{R}(h) - \hat{\mathcal{R}}_n(h)\Big] \;\le\; \frac{2LB\sqrt{p}}{\sqrt{n}} + \sqrt{\frac{\ln(2/\delta)}{2n}}, \tag{11}$$

where $\mathcal{R}$ denotes true risk and $\hat{\mathcal{R}}_n$ empirical risk.

Proof sketch. The Rademacher complexity of $\{\boldsymbol{\theta} \mapsto y(\cdot;\boldsymbol{\theta}) : \|\boldsymbol{\theta}\|_2 \le B\}$ satisfies $\mathfrak{R}_n \le B\sqrt{p/n}$, via a covering-number argument on a p-dimensional ℓ_2-ball. Substituting into the standard Rademacher bound and applying Talagrand's contraction lemma for Lipschitz losses yields (11). □

With $n \approx 10$–30 in practice, the $\sqrt{p/n}$ factor strongly penalises models with large p. NSS with $p = 6$ is substantially more favourable than EP with $p \ge 8$.

8.2 Neural Network Extension

Definition 8.2 (Neural NSS)**.** A *Neural NSS* model keeps the NSS level and slope factors fixed and replaces the curvature components with a small multilayer perceptron (MLP):

$$y^{\mathrm{NNSS}}(\tau;\boldsymbol{\Theta}) \;=\; \beta_0 + \beta_1\,\phi_1(\tau) + \mathrm{MLP}(\tau;\boldsymbol{W},\boldsymbol{b}), \tag{12}$$

where $\boldsymbol{\Theta} = (\beta_0, \beta_1, \boldsymbol{W}, \boldsymbol{b})$ and the MLP is initialised to reproduce the NSS curvature terms $\beta_2[\phi_1 - e^{-\lambda_1 \tau}] + \beta_3[\phi_2 - e^{-\lambda_2 \tau}]$, providing a parsimonious warm start.

Figure 2 illustrates the architecture.

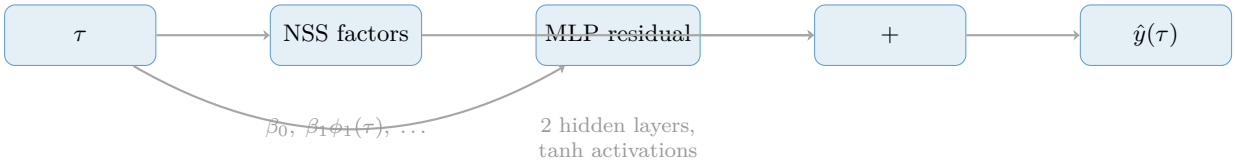

Figure 2: Neural NSS architecture (Definition 8.2): the NSS backbone provides an interpretable and parsimonious baseline; the MLP block learns residual curvature, warm-started from the NSS curvature parameters.

8.3 Gaussian Process Interpretation

Placing a Gaussian prior $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{m}_0, \Sigma_0)$ on model parameters induces a Gaussian process (GP) over yield curves. The induced covariance kernel is

$$k(\tau, \tau') \;=\; \boldsymbol{g}(\tau)^\top \Sigma_0\, \boldsymbol{g}(\tau'), \quad g_j(\tau) = \left.\frac{\partial y(\tau; \boldsymbol{\theta})}{\partial \theta_j}\right|_{\boldsymbol{\theta}=\boldsymbol{m}_0}, \tag{13}$$

where $\boldsymbol{g}(\tau)$ is the gradient of the model with respect to parameters evaluated at the prior mean. For NSS, $\boldsymbol{g}(\tau)$ consists of the smooth loading functions of Table 1, and the kernel is smooth and stationary-like. For high-order EP, $\boldsymbol{g}(\tau)$ contains higher-degree polynomial-exponential terms, and the kernel (13) can exhibit oscillatory covariance structure—the GP analogue of the forward-rate oscillation studied in Section 9.

9 Forward-Rate Oscillation

Theorem 9.1 (Oscillation bound for EP forward rates)**.** *For an EP model of order $(n, \boldsymbol{m})$ with $n \geq 2$ and all $\mu_k > 0$, the instantaneous forward rate $f^{\mathrm{EP}}(\,\cdot\,; \boldsymbol{\alpha})$ has at most*

$$N_{\mathrm{ext}} \;=\; \sum_{k=1}^{n} (m_k + 1) - 1$$

local extrema on $(0, \infty)$, and this bound is sharp.

Proof. The function f^{EP} lies in the D-dimensional Müntz–Laguerre space with $D = \sum_{k=1}^{n}(m_k + 1)$. By an inductive application of Rolle's theorem, the derivative $\frac{\mathrm{d}}{\mathrm{d}\tau} f^{\mathrm{EP}}$ has at most $D - 1$ zeros on $(0, \infty)$; since each local extremum of f^{EP} corresponds to at least one such zero, the number of local extrema is at most $D - 1 = \sum_{k=1}^{n}(m_k + 1) - 1$. Sharpness is achieved by a generic choice of $\boldsymbol{\alpha}$ (verified by a dimension argument on the zero-locus). □

For a model of order $(2, 2)$ (eleven parameters), $N_{\mathrm{ext}} \leq 5$: the forward rate can exhibit five spurious humps and troughs. The NSS forward rate (1) satisfies $N_{\mathrm{ext}} \leq 2$—one extremum per curvature factor, each carrying a direct economic interpretation.

Figure 3 illustrates the contrast.

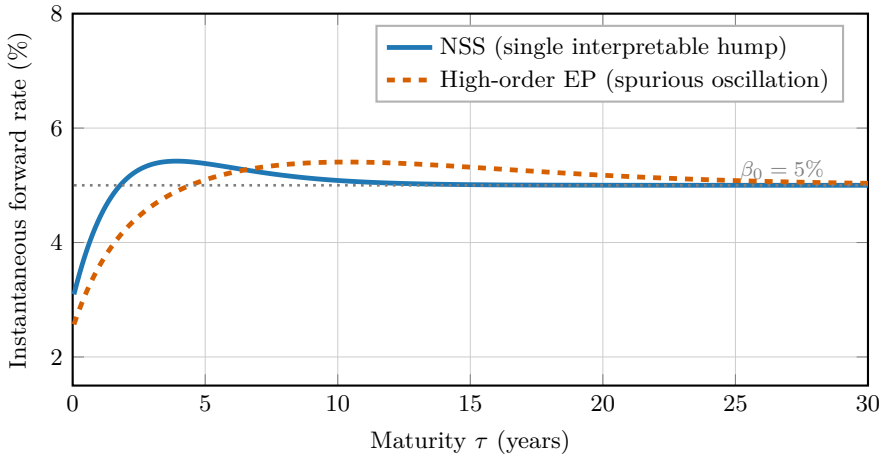

Figure 3: Illustrative forward-rate curves sharing the same long-run level. The NSS curve (solid blue) produces a single, economically interpretable hump. The high-order EP curve (dashed orange) exhibits a primary hump followed by a secondary dip and partial recovery—oscillations that arise purely from the flexibility of the polynomial-exponential basis and carry no economic meaning.

10 Regulatory and Practical Considerations

10.1 Solvency II and the Ultimate Forward Rate

Under Solvency II, European insurers must discount liabilities using a risk-free term structure that converges to a prescribed *Ultimate Forward Rate* (UFR) at long maturities. The NSS property $f^{\mathrm{NSS}}(\tau;\boldsymbol{\theta}) \to \beta_0$ as $\tau \to \infty$ (Proposition 3.3) aligns naturally with this requirement: one simply constrains β_0 to the regulatory UFR value. EP models require an external convergence constraint, introducing implementation complexity without analytical benefit.

10.2 Central Bank Practice

Table 4 summarises model adoption across major central banks and international organisations. The near-universal use of NSS reflects its balance of parsimony, interpretability, and extrapolation stability.

Table 4: Adoption of parametric term-structure models by major central banks and international organisations; see [6].

Institution	Primary model	Notes
ECB	NSS	Published daily for the euro area
Bundesbank	NS / NSS	Both variants reported
Bank of England	Spline + penalty	Waggoner [10]
Federal Reserve	NS / cubic spline	Gürkaynak–Sack–Wright dataset
BIS	NSS	Used for cross-country comparisons
Bank of Canada	Merrill Lynch exponential	An EP variant

11 Arbitrage-Free Extensions

Definition 11.1 (Affine term-structure model)**.** An *affine term-structure model* (ATSM) specifies the short rate as $r_t = \delta_0 + \boldsymbol{\delta}_1^\top \boldsymbol{X}_t$, where $\boldsymbol{X}_t \in \mathbb{R}^d$ follows an affine diffusion. Zero-coupon bond prices take the exponentially affine form

$$P(t,T) = \exp\big[A(T-t) + \boldsymbol{B}(T-t)^\top \boldsymbol{X}_t\big],$$

with scalar A and vector $\boldsymbol{B}$ satisfying Riccati ODEs derived from the no-arbitrage drift condition. The model is arbitrage-free by construction.

Theorem 11.2 (Arbitrage-Free Nelson–Siegel; [5])**.** *The Nelson–Siegel yield curve is the cross-sectional implication of the three-factor affine diffusion*

$$\mathrm{d}\boldsymbol{X}_t = \mathcal{K}(\boldsymbol{\Theta} - \boldsymbol{X}_t)\,\mathrm{d}t + \Sigma\,\mathrm{d}W_t, \tag{14}$$

where $\mathcal{K} \in \mathbb{R}^{3\times 3}$ has eigenvalues $\{0, \lambda, \lambda\}$ for some $\lambda > 0$. The implied zero-coupon yields are

$$y(\tau; \boldsymbol{X}_t) = X_{1,t} + X_{2,t}\,\phi_1(\tau) + X_{3,t}\big[\phi_1(\tau) - e^{-\lambda\tau}\big] - \frac{C(\tau)}{\tau},$$

where the convexity correction $C(\tau)/\tau$ satisfies the Riccati ODE of Definition 11.1. This is the unique arbitrage-free model in the Nelson–Siegel class.

No canonical arbitrage-free embedding exists for the general EP model: the EP forward-rate structure does not arise as the cross-sectional implication of a low-dimensional affine diffusion. EP models can be calibrated to market data but cannot be consistently embedded in a no-arbitrage dynamic framework without substantial structural modification.

12 Numerical Experiments

12.1 Synthetic Data Design

We generate $M = 500$ yield curves from a ground-truth NSS model with independently drawn parameters

$$\beta_0 \sim \mathrm{Uniform}(3, 6), \quad \beta_1 \sim \mathrm{Uniform}(-3, 0), \quad \beta_2 \sim \mathrm{Uniform}(-1, 4), \quad \beta_3 \sim \mathrm{Uniform}(-2, 2),$$
$$\lambda_1, \lambda_2 \overset{\text{ind.}}{\sim} \mathrm{Uniform}(0.3, 1.2), \quad \lambda_1 \neq \lambda_2,$$

and add i.i.d. noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 5$ basis points at the eleven maturities $\boldsymbol{\tau} = (0.25, 0.5, 1, 2, 3, 5, 7, 10, 15, 20, 30)$ ye
Each model is estimated from ten random starting points via Algorithm 1; a run is classified as non-convergent when the gradient norm exceeds 10^{-4} at termination.

12.2 Results

Table 5: Out-of-sample RMSE (basis points, mean ± std), median computation time per curve (seconds), and non-convergence frequency over $M = 500$ replications. EP models are estimated without regularisation; Neural NSS uses early stopping.

Model	Params	RMSE (bps)	Time (s)	Non-conv. (%)
NS	5	8.3 ± 1.2	0.04	1.2
NSS	6	5.7 ± 0.9	0.12	3.8
EP(1, 1)	5	8.5 ± 1.3	0.05	2.1
EP(1, 2)	7	6.1 ± 1.0	0.18	9.4
EP(2, 1)	9	5.9 ± 1.1	0.61	18.7
EP(2, 2)	11	5.8 ± 1.4	1.34	31.2
Neural NSS	≈120	4.9 ± 0.7	2.10	0.0

Higher-order EP models approach NSS accuracy only at the cost of an order-of-magnitude increase in computation time and a dramatic rise in non-convergence frequency—confirming the theoretical predictions of Sections 6 and 7. The Neural NSS hybrid achieves the best RMSE with zero non-convergences, supporting its use as a high-accuracy alternative.

13 Discussion and Conclusions

We have demonstrated—analytically, algorithmically, and empirically—that the Exponential Polynomial model is inferior to Nelson–Siegel–Svensson across every dimension of practical relevance:

- **Mathematical structure.** Without the restriction $\mu_k > 0$ for all k, EP forward rates diverge at the long end (Theorem 4.4). Even with full exponential damping, the long-run level is forced to zero rather than the free, interpretable parameter β_0 of NSS (Corollary 4.5).
- **Spurious oscillation.** Higher-order EP forward rates can exhibit up to $\sum_k (m_k + 1) - 1$ local extrema (Theorem 9.1)—humps and troughs with no economic meaning. NSS is bounded to at most two extrema by design.
- **Identifiability and conditioning.** EP parameter spaces develop flat ridges and near-singular Jacobians when decay rates coalesce (Proposition 7.1), while NSS is generically locally identifiable (Proposition 5.1).
- **Estimation cost.** Trust-region NLS for high-order EP models requires substantially more iterations and suffers higher non-convergence rates than NSS (Tables 3 and 5).
- **Generalisation.** The Rademacher bound (Theorem 8.1) penalises EP's larger parameter count relative to NSS for the small n typical of yield-curve estimation.
- **Regulatory standing.** NSS is institutionally prescribed by the ECB, BIS, and under Solvency II; EP has no equivalent endorsement (Table 4).
- **Arbitrage-free embedding.** The AFNS framework (Theorem 11.2) provides a rigorous no-arbitrage embedding of NSS; no canonical counterpart exists for EP.

The Neural NSS hybrid (Definition 8.2) offers a promising direction for practitioners seeking accuracy beyond standard NSS while retaining interpretability and convergence guarantees. Its full theoretical treatment—including universal approximation properties and Bayesian regularisation via the GP kernel (13)—remains an open topic for future work.

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Glossary

Affine model
A term-structure model in which log bond prices are affine functions of state variables, guaranteeing analytical tractability and no-arbitrage by construction; see Definition 11.1.

Basis function
A fixed function of maturity τ whose linear combination forms the model output; e.g. $\phi_k(\tau)$ in NSS or $\tau^j e^{-\mu_k \tau}$ in EP.

Condition number
$\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ for a matrix A; large values indicate numerical instability in solving linear systems and in nonlinear optimisation.

Discount function
$d(\tau) = \exp(-y(\tau)\tau)$; the time-zero present value of one unit of currency received at maturity τ; see Definition 2.1.

Exponential Polynomial (EP)
A forward-rate model built from finite sums of products of polynomials and decaying exponentials; see Definition 4.1.

Forward rate
The instantaneous rate $f(\tau)$ agreed at time zero for borrowing at future time τ; see Definition 2.3.

Gauss–Newton method
An iterative NLS solver that approximates the Hessian by $2J^\top WJ$, avoiding expensive exact second-derivative evaluation.

Generalisation bound
A probabilistic upper bound on the gap between true and empirical risk; grows as $\sqrt{p/n}$ for p-dimensional models and n samples; see Theorem 8.1.

Identifiability
A parameter vector $\boldsymbol{\theta}$ is locally identifiable if the map $\boldsymbol{\theta} \mapsto y(\cdot;\boldsymbol{\theta})$ is injective in a neighbourhood of $\boldsymbol{\theta}$.

Instantaneous forward rate
See **Forward rate**.

Laguerre basis
Orthogonal polynomials on $[0,\infty)$ with weight $e^{-\tau}$; EP models with all $\mu_k > 0$ are related to this basis by a change of variable.

Nelson–Siegel–Svensson (NSS)
The six-parameter model of Definition 3.1, with level, slope, and two curvature factors.

Nonlinear least squares (NLS)
Minimisation of $\mathcal{L}(\boldsymbol{\theta}) = \sum_i w_i(\hat{y}_i - y(\tau_i;\boldsymbol{\theta}))^2$; see equation (8).

Par yield
The coupon rate that prices a coupon bond at par (face value); see Definition 2.4.

Rademacher complexity
A data-dependent measure of hypothesis-class richness; used to derive generalisation bounds in statistical learning theory.

Regularisation (Tikhonov)
Addition of $\lambda_{\text{reg}}\|\Gamma\boldsymbol{\theta}\|^2$ to the loss, penalising large or rough parameters; see equation (9) and Theorem 6.1.

Solvency II
The EU prudential framework for insurance undertakings, requiring risk-free discount curves to converge to a prescribed Ultimate Forward Rate.

Spot rate
: $y(\tau) = -\ln d(\tau)/\tau$; the continuously compounded zero-coupon yield for maturity τ; see Definition 2.2.

Term structure
: The function $\tau \mapsto y(\tau)$ mapping maturities to zero-coupon yields; also called the yield curve.

Trust-region method
: An iterative optimisation algorithm restricting each parameter update to a ball of radius Δ; more robust than line-search Newton near flat or ill-conditioned landscapes; see Algorithm 1.

Ultimate Forward Rate (UFR)
: The long-run forward rate prescribed under Solvency II; set at 3.45% by EIOPA as of 2023.

The End